

```

# Vector

v <- as.numeric(strsplit(
  readline("Enter 3 vector values: "), " ")
)[[1]])

# Matrix

m_values <- as.numeric(strsplit(
  readline("Enter 4 matrix values: "), " ")
)[[1]])

m <- matrix(m_values, nrow = 2, ncol = 2)

# Function

square <- function(x) {
  return(x * x)
}

# Create list

mylist <- list(
  Vector = v,
  Matrix = m,
  Function = square
)

cat("\nComplete List:\n")

print(mylist)

x <- as.numeric(readline("\nEnter a number for function: "))

cat("Square =", mylist$Function(x))

```

RStudio interface showing an R script, environment, and console.

**Script Editor (EX1.R):**

```
3 readline("Enter 3 vector values: "), " "  
4 >[[1]]  
5 # Matrix  
6 m_values <- as.numeric(strsplit(  
7 readline("Enter 4 matrix values: "), " "  
8 >[[1]])  
9 m <- matrix(m_values, nrow = 2, ncol = 2)  
10 # Function  
11 square <- function(x) {  
12   return(x * x)  
13 }  
14 # Create list  
15 mylist <- list(  
16   vector = v,  
17   Matrix = m,  
18   Function = square  
19 )  
20 cat("\nComplete List:\n")  
21 print(mylist)  
22 x <- as.numeric(readline("\nEnter a number for function: "))  
23 cat("Square =", mylist$Function(x))  
24
```

**Environment:**

| Object           | Class          | Value    |
|------------------|----------------|----------|
| m                | num [1:2, 1:2] | 8 6 4 3  |
| mylist           | List of 3      |          |
| <b>Values</b>    |                |          |
| m_values         | num [1:4]      | 8 6 4 3  |
| v                | num [1:3]      | 10 55 99 |
| x                | num            | 8        |
| <b>Functions</b> |                |          |
| square           | function (x)   |          |

**Console:**

```
> source("C:/Users/2021e/Desktop/EX 1.R")  
Enter 3 vector values: 10 55 99  
Enter 4 matrix values: 8 6 4 3  
  
Complete List:  
$vector  
[1] 10 55 99  
  
$Matrix  
      [,1] [,2]  
[1,]    8    4  
[2,]    6    3  
  
$Function  
function(x) {  
  return(x * x)  
}  
  
Enter a number for function: 8  
Square = 64  
>
```